📘 Chapter 23- CDP, LLDP and NTP (Network Time Protocol)

🌐 Part 1: Layer 2 Discovery Protocols

📌 What are Layer 2 Discovery Protocols?

Layer 2 discovery protocols are used by network devices to **discover information about directly connected neighbor devices**.

They work at the **Data Link Layer (Layer 2)** of the OSI Model.

Devices share information such as:

- 💻 Hostname
- 🌍 IP Address
- 🔧 Device Type
- 📦 Platform/Model
- 🔌 Interface information
- 🏷️ VLAN information

Two important Layer 2 discovery protocols:

Protocol	Type	Vendor
CDP	Proprietary	Cisco
LLDP	Open Standard	IEEE

📡 Why Are Discovery Protocols Useful?

They help network engineers understand **what devices are connected in the network**.

Example 🏢

Imagine a large office network.

A network engineer connects to a switch and wants to know:

- What router is connected?
- What interface is connected?
- What device type is it?

Instead of checking cables manually, **CDP or LLDP automatically shows the information**.

So the engineer can easily see the network topology.

⚠️ Security Risk of CDP and LLDP

Both protocols **broadcast device information** in the network.

Information sent includes:

- Hostname
- IP address
- Software version
- Interface details
- VLAN information

If a malicious device connects to the network, it can **capture this information**.

Possible attacks:

- Network mapping
- Targeted attacks
- Man-in-the-Middle attacks

Example 🔐

Imagine a **rogue laptop connected to an unused office switch port**.

Using CDP or LLDP packets, the attacker can learn:

- Router IP addresses
- Device models

- Software versions

This helps them plan attacks.

Because of this, these protocols are **often disabled in secure environments**.

🖧 Cisco Discovery Protocol (CDP)

📌 What is CDP?

CDP is a **Cisco proprietary protocol** used to discover information about **neighboring Cisco devices**.

Only **Cisco devices can understand CDP messages**.

Devices that support CDP:

- Cisco routers
- Cisco switches
- Cisco IP phones
- Cisco firewalls

⚙️ CDP Default Behavior

By default:

- CDP is **enabled globally**
- CDP is **enabled on all interfaces**

CDP sends messages every:

⏱ **60 seconds**

CDP holdtime:

⏱ **180 seconds**

If no message is received within **180 seconds**, the device is removed from the **CDP neighbor table**.

📡 CDP Multicast Address

CDP messages are sent to:

0100.0CCC.CCCC

When a device receives the message:

1️⃣ It processes the message
2️⃣ Stores information in the neighbor table
3️⃣ Discards the message

It **does NOT forward the message** to other devices.

📋 CDP Neighbor Table

You can view neighbors using:

show cdp neighbors

This command displays information such as:

Field	Meaning
Device ID	Name of neighbor device
Local Interface	Interface on local device
Holdtime	Time before entry expires
Capability	Device type
Platform	Device model
Port ID	Neighbor interface

🔎 Detailed CDP Information

Command:

show cdp neighbors detail

Shows additional details such as:

- IOS version

- IP address
- Device platform
- Interface details

⚙️ CDP Configuration Commands

Enable CDP Globally

R1(config)# cdp run

Disable CDP Globally

R1(config)# no cdp run

Enable CDP on Interface

R1(config)# interface g0/1
R1(config-if)# cdp enable

Disable CDP on Interface

R1(config)# interface g0/1
R1(config-if)# no cdp enable

Configure CDP Timer

R1(config)# cdp timer 60

Configure CDP Holdtime

R1(config)# cdp holdtime 180

Enable/Disable CDPv2

R1(config)# cdp advertise-v2

Disable:

R1(config)# no cdp advertise-v2

🌍 Link Layer Discovery Protocol (LLDP)

📌 What is LLDP?

LLDP is a **vendor-neutral discovery protocol**.

It works between **different vendors**.

Example vendors:

- Cisco
- Juniper
- HP
- Dell

LLDP standard:

IEEE 802.1AB

Example Scenario 🏢

Suppose a network has:

- Cisco Router
- HP Switch
- Juniper Firewall

CDP will **not work between them**.

But LLDP will work because it is an **open standard protocol**.

LLDP Default Behavior

On Cisco devices:

- LLDP is **disabled by default**

It must be enabled manually.

LLDP Multicast Address

LLDP messages are sent to:

0180.c200.000E

LLDP Timers

Timer	Value
Message timer	30 seconds
Holdtime	120 seconds
Reinitialization delay	2 seconds

⚙️ LLDP Configuration Commands

Enable LLDP Globally

R1(config)# lldp run

Enable LLDP Transmission

R1(config-if)# lldp transmit

Enable LLDP Reception

R1(config-if)# lldp receive

Both **transmit and receive must be enabled** for full communication.

Disable LLDP

R1(config)# no lldp run

🔎 LLDP Verification Commands

Show neighbors:

show lldp neighbors

Detailed output:

show lldp neighbors detail

Show LLDP status:

show lldp

📊 CDP vs LLDP Comparison

Feature	CDP	LLDP
Vendor	Cisco	IEEE Standard
Compatibility	Cisco only	Multi-vendor
Default State	Enabled	Disabled
Layer	Layer 2	Layer 2
Use Case	Cisco networks	Mixed networks

⏰ Part 2: Network Time Protocol (NTP)

📌 What is NTP?

NTP stands for:

Network Time Protocol

It is used to **synchronize time across all devices in a network**.

Devices include:

- Routers
- Switches
- Servers
- PCs

🌍 Daily Life Example

Imagine three routers:

- Router 1 time → 10:00 AM
- Router 2 time → 10:05 AM
- Router 3 time → 9:58 AM

If a network failure happens, logs will show **different timestamps**.

The engineer cannot determine:

- Which event happened first
- Which device failed first

Using **NTP**, all routers follow the **same time**.

So troubleshooting becomes easier.

🔐 Why Time Synchronization is Important

Reason	Explanation
Logs	Accurate timestamps
Security	Certificates need correct time
Troubleshooting	Events occur in correct order
Auditing	Legal logs require accuracy
Automation	Scheduled tasks run correctly

🧠 NTP Key Concepts

Term	Meaning
NTP Server	Device providing time
NTP Client	Device receiving time
Stratum	Distance from reference clock
NTP Master	Router acting as time server
NTP Associations	Shows connected time servers

🧭 Stratum Levels

Stratum	Description
0	Atomic clock / GPS
1	Connected to stratum 0
2	Gets time from stratum 1
3–15	Lower accuracy
16	Unsynchronized

Lower stratum = **higher accuracy**.

⏲️ Types of Clocks in Routers

1️⃣ Software Clock

- Set manually
- Temporary
- May reset after reboot

2️⃣ Hardware Clock (Calendar)

- Stored in device hardware
- Survives reboot
- Used during startup

🔎 Clock Commands

Show current time:

show clock

Detailed clock information:

show clock detail

⌨️ Manually Setting Time

clock set 11:50:00 Mar 5 2026

This sets the **software clock only**.

🔋 Hardware Clock Commands

Update calendar from clock:

clock update-calendar

Read calendar to clock:

clock read-calendar

🌍 Time Zone Configuration

Example:

clock timezone IST 5 30

This sets **Indian Standard Time**.

🗄 NTP Port Number

NTP uses:

UDP Port 123

🧪 CCNA NTP Lab (3 Router Setup)

Network Topology

R2 ---- R1 ---- R3

Networks:

R1-R2 → 192.168.20.0
R1-R3 → 192.168.10.0

R1 will act as **NTP Master**.

Step 1: Configure IP Addressing

Router 1

interface g0/0
ip address 192.168.20.1 255.255.255.0
no shutdown

interface g0/1
ip address 192.168.10.1 255.255.255.0
no shutdown

Router 2

hostname R2

interface g0/0
ip address 192.168.20.2 255.255.255.0
no shutdown

Router 3

hostname R3

interface g0/0
ip address 192.168.10.2 255.255.255.0
no shutdown

Step 2: Configure R1 as NTP Master

clock timezone IST 5 30
clock set 11:50:00 Mar 5 2026
ntp master 1

Stratum level = **1**

Step 3: Configure NTP Clients

Router 2

ntp server 192.168.20.1

Router 3

ntp server 192.168.10.1

Step 4: Verification Commands

Run on R2 and R3:

show ntp associations
show ntp status
show clock

✅ Expected Result

- Clock synchronized ✓
- Time source = R1 ✓
- Same time on all routers ✓

📚 Important CCNA Exam Points

✓ CDP is Cisco proprietary
✓ LLDP is open standard (IEEE 802.1AB)
✓ NTP uses UDP port 123
✓ Stratum indicates distance from reference clock
✓ Discovery protocols operate at Layer 2
✓ CDP default timer = 60 seconds
✓ CDP holdtime = 180 seconds

📚 CHAPTER SUMMARY

🌐 1️⃣ Layer 2 Discovery Protocols

Layer 2 discovery protocols allow network devices to **discover information about directly connected neighbors**.

They operate at **OSI Layer 2 (Data Link Layer)**.

Devices share information like:

- 🖥️ Hostname
- 🌍 IP Address
- 🔧 Device Type
- 📦 Hardware Model
- 🔌 Interface Port
- 🏷️ VLAN Information
- ⚙️ Software Version

These protocols help engineers **visualize network topology without physically checking cables**.

📡 Why Discovery Protocols Are Important

In large enterprise networks, engineers must know:

- Which device is connected
- Which interface is used
- What device type it is
- What operating system version it runs

Instead of tracing cables manually, **discovery protocols automatically provide this information**.

Example 🏢
A network engineer logs into a switch and instantly sees:

Router connected on G0/1
Switch connected on G0/2
IP Phone connected on Fa0/10

This greatly simplifies **troubleshooting and network documentation**.

⚠️ Security Risk of Discovery Protocols

Both **CDP and LLDP broadcast device information**.

Information exposed includes:

- Hostname
- Device model
- Software version
- IP addresses
- VLAN information
- Interface details

🚨 If a malicious device connects to the network, it can capture these packets.

Possible attacks:

- Network reconnaissance
- Targeted vulnerability attacks
- Man-in-the-Middle attack planning

Because of this risk, **many organizations disable discovery protocols on external interfaces**.

🖧 2️⃣ Cisco Discovery Protocol (CDP)

📌 What is CDP?

CDP is a **Cisco proprietary Layer 2 protocol** used to discover information about neighboring Cisco devices.

Only **Cisco devices can understand CDP messages**.

Supported devices include:

- Cisco routers
- Cisco switches
- Cisco IP phones
- Cisco firewalls

⚙️ Default Behavior of CDP

By default:

- CDP is **enabled globally**
- CDP runs on **all interfaces**

Timers

Setting	Value
Message timer	60 seconds
Holdtime	180 seconds

If a device does not receive CDP updates within **180 seconds**, the neighbor entry is removed.

📡 CDP Multicast Address

CDP uses the multicast MAC address:

0100.0CCC.CCCC

When a device receives CDP packet:

1️⃣ Processes information
2️⃣ Stores in neighbor table
3️⃣ Discards packet

It **does NOT forward it further**.

📋 CDP Neighbor Table

Command:

show cdp neighbors

Output includes:

Field	**Meaning**
Device ID	Neighbor hostname
Local Interface	Interface on local device
Holdtime	Time remaining
Capability	Device type
Platform	Hardware model
Port ID	Neighbor interface

🔎 Detailed CDP Information

Command:

show cdp neighbors detail

Shows additional data:

- IP address
- IOS version
- Platform model
- Interface details

⚙️ CDP Configuration Commands

Enable CDP globally

cdp run

Disable CDP globally

no cdp run

Enable on interface

interface g0/1
cdp enable

Disable on interface

interface g0/1
no cdp enable

🌍 3️⃣ Link Layer Discovery Protocol (LLDP)

📌 What is LLDP?

LLDP is a **vendor-neutral Layer 2 discovery protocol**.

It is defined by:

IEEE 802.1AB

LLDP allows devices from **different vendors to communicate**.

Example vendors:

- Cisco
- Juniper
- HP
- Dell
- Arista

Example Network

Cisco Router
⬇
HP Switch
⬇
Juniper Firewall

CDP will **not work across vendors**, but LLDP **works perfectly**.

LLDP Default Behavior

On Cisco devices:

❌ LLDP is disabled by default.

It must be enabled manually.

LLDP Multicast Address

0180.c200.000E

LLDP Timers

Parameter	Value
Message timer	30 seconds
Holdtime	120 seconds
Reinit delay	2 seconds

⚙️ **LLDP Configuration Commands**

Enable globally

lldp run

Enable sending packets

lldp transmit

Enable receiving packets

lldp receive

Disable LLDP

no lldp run

🔎 LLDP Verification

show lldp neighbors
show lldp neighbors detail
show lldp

📊 CDP vs LLDP

Feature	CDP	LLDP
Vendor	Cisco	IEEE
Compatibility	Cisco only	Multi-vendor
Default state	Enabled	Disabled
Layer	Layer 2	Layer 2
Usage	Cisco networks	Mixed vendor networks

⏰ 4️⃣ Network Time Protocol (NTP)

📌 What is NTP?

NTP stands for **Network Time Protocol**.

It synchronizes **time across all network devices**.

Devices include:

- Routers
- Switches
- Servers

- PCs

🌍 Why Time Synchronization Is Important

Reason	Explanation
Logs	Accurate timestamps
Security	Certificates require correct time
Troubleshooting	Event order must be correct
Automation	Scheduled jobs run correctly
Compliance	Legal audit logs require accuracy

🧠 NTP Key Concepts

Term	Meaning
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NTP Associations	Shows connected servers

🧭 Stratum Levels

Stratum	Description
0	Atomic clock / GPS
1	Connected to stratum 0
2	Gets time from stratum 1
3–15	Lower accuracy

Stratum	Description
16	Unsynchronized

Lower stratum = **more accurate clock**.

🕛 Router Clocks

1️⃣ Software Clock

- Set manually
- Temporary
- Lost after reboot

2️⃣ Hardware Clock (Calendar)

- Stored in hardware
- Survives reboot
- Used during startup

🔎 Clock Commands

Show current time

show clock

Detailed clock

show clock detail

⌨️ Manually Setting Time

clock set 11:50:00 Mar 5 2026

Sets **software clock only**.

🔋 Hardware Clock Commands

Update calendar

clock update-calendar

Read calendar

clock read-calendar

🌍 Timezone Configuration

Example:

clock timezone IST 5 30

Sets Indian Standard Time.

🚪 NTP Port Number

NTP uses:

UDP Port 123

🎯 CHAPTER CONCLUSION

This chapter explains **two major networking functions**:

🔎 Network Discovery

Using **CDP and LLDP**, engineers can automatically detect:

- Connected devices
- Interfaces
- IP addresses
- Device models

⏰ Time Synchronization

Using **NTP**, all devices maintain **accurate and consistent system time**.

These technologies are essential for:

- Troubleshooting
- Monitoring

- Security
- Network documentation

🎤 Q & A

1️⃣ What is CDP?

CDP (Cisco Discovery Protocol) is a Layer 2 proprietary protocol developed by Cisco that allows Cisco devices to discover information about directly connected Cisco neighbors.

It provides information like:

- Device hostname
- IP address
- Platform model
- Interface details

This helps engineers understand the network topology.

2️⃣ What is LLDP?

LLDP is an open standard Layer 2 discovery protocol defined by IEEE 802.1AB.

Unlike CDP, LLDP works between devices from different vendors such as Cisco, HP, Juniper, and Dell.

It allows network devices to advertise their identity and capabilities.

3️⃣ Difference between CDP and LLDP?

Feature	CDP	LLDP
Type	Proprietary	Open standard
Vendor support	Cisco only	Multi-vendor

Feature	CDP	LLDP
Default state	Enabled	Disabled
Standard	Cisco	IEEE 802.1AB

4️⃣ At which OSI layer do CDP and LLDP operate?

They operate at:

OSI Layer 2 (Data Link Layer).

They do not rely on IP addresses because they operate using **MAC addresses**.

5️⃣ What information does CDP provide?

CDP provides several details about neighboring devices:

- Hostname
- IP address
- Device type
- Interface information
- Software version
- Platform model

6️⃣ What is CDP holdtime?

Holdtime is the duration that a CDP neighbor entry remains in the table without receiving updates.

Default value:

180 seconds

If updates stop for 180 seconds, the entry is removed.

7️⃣ What is CDP timer?

CDP timer determines how often CDP messages are sent.

Default value:

60 seconds

8️⃣ What multicast address does CDP use?

CDP sends packets to:

0100.0CCC.CCCC

9️⃣ What multicast address does LLDP use?

LLDP sends packets to:

0180.c200.000E

🔟 What commands verify CDP neighbors?

show cdp neighbors
show cdp neighbors detail

11️⃣ Why might CDP be disabled?

CDP may expose sensitive network information like:

- Device models
- IP addresses
- Software versions

Attackers could use this information for reconnaissance.

Therefore it is often disabled on **internet-facing interfaces**.

1️⃣2️⃣ What is NTP?

NTP (Network Time Protocol) synchronizes the system clock of network devices using a hierarchical system of time servers.

It ensures that all devices have **consistent timestamps**.

1️⃣3️⃣ What port does NTP use?

NTP uses:

UDP Port 123

1️⃣4️⃣ What is NTP stratum?

Stratum indicates how far a device is from the reference clock.

Lower stratum means higher accuracy.

Example:

- Stratum 1 → very accurate
- Stratum 3 → less accurate

1️⃣5️⃣ What is NTP master?

An NTP master is a device configured to act as a **time source for other devices**.

Example command:

ntp master 1

🧪 PRACTICAL INTERVIEW QUESTIONS

1️⃣6️⃣ How do you configure NTP client?

ntp server 192.168.1.1

This tells the router to synchronize time with that server.

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1 7️⃣ How do you check NTP status?

show ntp status

Shows synchronization state.

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1 8️⃣ How do you check NTP associations?

show ntp associations

Shows connected NTP servers.

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1 9️⃣ What happens if time is not synchronized?

Several issues may occur:

- Incorrect logs
- Security certificate failures
- Incorrect event ordering
- Automation failures

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2 0️⃣ How do you disable CDP?

Globally:

no cdp run

On interface:

no cdp enable
